

ACCELERATOR DESIGN EDUCATIONAL PRIMER – CONCEPTUALIZING AND OPTIMIZING THE HYBRID LHEC-LIKE ELECTRON-ION COLLIDER DESIGN*

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Abstract

The Electron-Ion Collider (EIC) Mission Need requires $\sqrt{s} = 20\text{--}100$ GeV (upgradable to 140 GeV) and luminosity $10^{33}\text{--}10^{34}$ cm⁻² s⁻¹. The current ring-ring baseline achieves the full scope, including $\sim 10^{34}$ cm⁻² s⁻¹ across all energies. However, when the design is re-optimized for the lower boundary — accepting 10^{33} cm⁻² s⁻¹ and prioritizing cost — an alternative configuration emerges as more advantageous: a hybrid LHeC-like electron accelerator using multi-pass energy recovery linacs (ERL). This solution reduces electron-beam power by roughly an order of magnitude, yielding nearly a factor of two reduction in total project cost compared with the present baseline while still satisfying the minimum physics requirements. The study performs parametric cost and performance modeling, augmented by AI-driven optimization, to explore this design space. Serving primarily as an educational exercise for the next generation of accelerator physicists and engineers, the paper demonstrates modern design methods: rapid parametric scans, cost-driven optimization, and integration of AI tools. It examines technical feasibility, identifies critical R&D (high-current ERL operation, beam-beam effects, synchronization, etc.), and discusses how such re-optimization studies can be used to train designers in an era when artificial intelligence dramatically expands exploration of complex accelerator parameter spaces.

PREAMBLE – AI AND EDUCATION

Educating future accelerator scientists in the AI era require developing new approaches. This paper is not an ultimate answer, but just a small step, hopefully in the right direction, along the path of finding optimal methodology of transferring knowledge to the new generation in the new world of unprecedented acceleration of analysis, simulations, and coding, due to proliferation of AI.

The emergence of AI-assisted coding presents both unique challenges and unprecedented opportunities for accelerator physics education. Institutions like Fermilab, which hosts the US Particle Accelerator School, and CERN, home to the CERN Accelerator School CAS, need to rapidly adapt their pedagogical approaches to this new reality. Launch of

the Genesis program in the US further encourages use of AI for accelerated design of particle accelerators.

In this paper, we will try to mimic what can be done in one-week USPAS course on accelerator design, where students typically analyze and collectively develop a mini project. Past examples of such USPAS mini-projects are compact LWFA-fed ring light source [USPAS2016] or ERL-based EUV light source [USPAS2023].

Those projects mentioned above were performed without use of AI. For the present paper, we use the same approach of USPAS mini-project, but assume that students will have access to AI tools, and will make most of design steps using AI. The authors of this paper are not students, however, we are trying to imagine what we could do in one week, if we were given the described below mini-project. In a sense, the paper is a rehearsal or a draft script for a future USPAS course, where accelerator mini-projects will be evaluated and conceptualized using AI tools.

EIC-LHEC MINI-PROJECT

The EIC-LHeC mini-project is conceived in the spirit of a one-week USPAS accelerator design course, where a small team of students is given a specific design challenge and expected to produce a self-consistent conceptual study by the end of the week. Past USPAS mini-projects — such as a compact LWFA-fed ring light source [USPAS2016] and an ERL-based EUV light source [USPAS2023] — were completed without AI assistance. The present work uses the same pedagogical framework, but now assumes that students have access to modern AI tools throughout every stage of the design process.

The mini-project assignment can be stated as follows: *Given the EIC proton beam stored in the RHIC-derived Hadron Storage Ring, design the simplest and most cost-effective electron accelerator complex that satisfies the EIC CD0 Mission Need lower boundary — a center-of-mass energy of $\sqrt{s} \approx 29\text{--}100$ GeV and an average luminosity of $\mathcal{L} \sim 10^{33}$ cm⁻² s⁻¹ — and estimate its cost relative to the ring-ring baseline.*

The motivation for this re-examination comes from discussions at the European Strategy for Particle Physics (ESPP) Open Symposium in July 2025 [ESPP], where it was argued that inserting a new electron storage ring into the LHC tunnel for LHeC would be both cost-prohibitive and luminosity-limiting. This observation prompted the question: could a

* Parameters of the presented concepts were developed with assistance from Grok AI and Claude AI.

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similar trade-off – replacing an electron storage ring with a superconducting energy-recovery linac (ERL) – be beneficial in the EIC context as well?

The proposed hybrid configuration keeps the full EIC proton beam infrastructure intact and replaces the electron storage ring with a multi-pass ERL, inspired by the LHeC linac-ring design [ESPP-LHeC-submission]. Electrons are accelerated in a superconducting RF (SRF) linac, transported through recirculation arcs, brought to collision, and then decelerated back through the linac, recovering most of the beam energy. This approach has several appealing features compared to the ring-ring baseline:

- The electron beam power is reduced by roughly an order of magnitude, since each electron bunch is used only once rather than stored for millions of turns.
- The absence of a large electron storage ring substantially reduces civil construction and magnet costs.
- A 6 GeV SRF linac traversed three times reaches 18 GeV, covering the full EIC energy range without additional hardware.
- The electron beam emittance can in principle be kept very small, since it is set at the injector and not blown up by synchrotron radiation in a ring.

The layout of this hybrid design is shown schematically in Fig. 1: a 6 GeV SRF ERL linac is installed across a diameter of the existing RHIC/EIC tunnel, with recirculation arcs using conventional (non-FFA) magnets routed around the outside of the ion ring. The electron beam is delivered to the interaction region, shared with the ePIC detector, and the spent beam is dumped after energy recovery.

The mini-project proceeds through the following design steps, each of which was carried out with AI assistance and is described in subsequent sections of this paper:

1. Parametric modeling of beam parameters, luminosity, and beam-beam effects for the baseline EIC, the reference LHeC design, and the proposed hybrid (*EIC-LHeC parameters* section).
2. Evaluation of beam-beam effects and electron disruption at the interaction point (*Beam-beam evaluations* section).
3. Conceptual IR optics design and estimation of final-doublet aperture requirements (*IR optics evaluations* section).
4. Assessment of IR magnet technology, including the choice between Nb–Ti and Nb₃Sn superconducting options (*IR magnets evaluations* section).
5. Feasibility analysis of energy recovery at high current and with a disrupted electron beam (*Energy recovery evaluations* section).

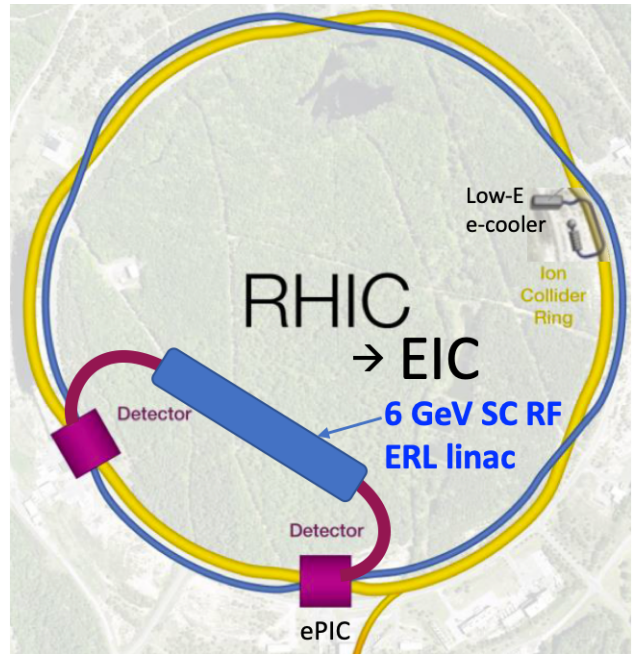


Figure 1: Schematic layout of the EIC built as an LHeC-like hybrid with energy recovery. A 6 GeV SC RF ERL linac (blue rectangle) is installed across a diameter of the existing RHIC/EIC tunnel. Electrons reach 10–18 GeV via two or three recirculation passes. The ePIC detector is retained at the existing interaction point.

6. Design of the recirculation arc optics and transition sections (*Recirculation evaluations* section).
7. Parametric cost estimation of the electron complex, interaction region, and detector, benchmarked against the LHeC cost study [LHeC-costs], and converted to USA DOE accounting (*Cost estimations* sections).
8. Identification of the critical R&D items and open questions (*Required R&D and next steps* section).

Throughout all these steps, AI tools – primarily large language models with code generation capability – were used to write Python scripts, interpret results, explore parameter trade-offs, and draft text. The role of the “students” (in this rehearsal, the authors) was to formulate physically meaningful prompts, critically verify the outputs against known accelerator physics principles, and iteratively refine both the prompts and the results. This workflow is described in detail in the *Verification of AI results* section.

EIC-LHEC PARAMETERS

Approach and AI Prompting Strategy

The parameter development for the EIC-LHeC hybrid followed an iterative dialogue with AI tools, beginning from the two parent designs and converging on a self-consistent hybrid configuration. The starting prompt sequence was designed to be accessible to a student with basic accelerator physics knowledge, progressively building up from simple

luminosity calculations to beam-beam constraints and cost-driven optimization.

The initial prompts requested Python codes to compute beam parameters and luminosity for the standard EIC and LHeC designs separately, using published parameter tables [EIC-pars, ESPP-LHeC-submission] as numerical inputs. The AI was asked to use normalized emittances, compute geometric emittances via the relativistic γ factor, calculate beam sizes at the interaction point (IP), and output a formatted parameter table. Beam-beam tune shift parameters $\xi_{x,y}$ and RMS beam divergences at the IP were added in subsequent prompts

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Resolving Parameter Inconsistencies in the Hybrid

When the AI was first asked to combine EIC proton parameters with LHeC electron parameters into a single hybrid collider, it correctly identified several fundamental inconsistencies between the two parent designs and flagged them before producing any code. The main conflicts were:

- **Bunch repetition rate mismatch.** The EIC ring circumference of 3834 m gives a revolution frequency of ~ 78 kHz and, with 1160 bunches, a collision frequency of ~ 91 MHz. The LHeC electron linac, matched to the LHC ring, assumes ~ 32 MHz. Naively combining the two would imply an electron beam current of ~ 1.45 A – far beyond the LHeC design value of 60 mA.
- **Beam size mismatch at the IP.** LHeC electron optics ($\beta_{x,y}^* = 0.20$ m, $\varepsilon_n = 22$ μm) were optimized to match the tiny ~ 10 μm proton beam sizes in the 7 TeV LHC. EIC protons at 275 GeV have much larger emittances, giving $\sigma_x \approx 173$ μm and $\sigma_y \approx 16$ μm at the LHeC β^* – a severe luminosity and beam-beam penalty.

- **Bunch population and current limits.** Keeping the LHeC electron bunch population of $N_e \approx 1.19 \times 10^{10}$ at the EIC collision frequency would require ~ 1.45 A electron current, violating the LHeC design limit. Conversely, enforcing 60 mA at the EIC frequency would require reducing N_e to $\sim 4 \times 10^9$.

The resolution adopted in the mini-project was to reduce the number of proton bunches by a factor of three relative to the EIC baseline (from 1160 to 386), proportionally increasing the bunch population while keeping the total proton beam current constant at 1.0 A. This brings the collision frequency down to ~ 30 MHz, closely matching the LHeC electron bunch rate and restoring a physically consistent electron beam current of ~ 58 mA. The IP optics were then re-optimized to match proton and electron beam sizes at the new operating point, yielding $\beta_x^* = 1.20$ m, $\beta_y^* = 0.05$ m for electrons and $\beta_x^* = 0.12$ m, $\beta_y^* = 0.06$ m for protons. This discussion with the AI also highlighted that the electron beam-beam parameter ξ_e is relatively unconstrained in the linac-ring scheme, since electrons are discarded after a single pass; the binding constraint is the *proton* beam-beam parameter, which was kept below $\xi_p < 0.01$.

A further optimization step explored paths to $\mathcal{L} = 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, relying on: (i) reduction of proton emittances via coherent electron cooling at collision energy; and (ii) reduction of the vertical electron emittance using a magnetized electron gun followed by Derbenev’s flat-beam transformer, conservatively assumed to reduce $\varepsilon_{n,y}^e$ by a factor of ten.

Baseline EIC and LHeC Reference Parameters

Tables 1 and 2 show abridged parameter sets for the two parent designs, retaining the quantities most relevant for luminosity and beam-beam evaluation. Full parameter tables are available in the references [EIC-pars, ESPP-LHeC-submission].

Table 1: Abridged Baseline EIC Parameters (1×10^{34} Peak Luminosity, 10 GeV e on 275 GeV p)

Parameter	Value	Unit
Electron / proton energy	10.0 / 275.0	GeV
$\sqrt{s}$	105	GeV
Number of bunches	1160	
Collision frequency	9.1×10^7	Hz
e bunch population	1.72×10^{11}	
p bunch population	6.9×10^{10}	
e norm. emittance h/v	391.0 / 26.0	μm
p norm. emittance h/v	3.3 / 0.3	μm
e $\beta_{x/y}^*$ at IP	0.45 / 0.06	m
p $\beta_{x/y}^*$ at IP	0.80 / 0.07	m
e $\sigma_{x/y}$ at IP	94.8 / 8.6	μm
p $\sigma_{x/y}$ at IP	94.9 / 8.6	μm
e / p beam current	1555 / 1000	mA
p beam-beam $\xi_{x/y}$	0.012 / 0.012	
Peak luminosity $\mathcal{L}$	1.05×10^{34}	$\text{cm}^{-2}\text{s}^{-1}$

Table 2: Abridged LHeC Reference Parameters (Phase-1, 20 GeV e on 7 TeV p)

Parameter	Value	Unit
Electron / proton energy	20.0 / 7000.0	GeV
$\sqrt{s}$	749	GeV
Number of bunches	2808	
Collision frequency	3.2×10^7	Hz
e bunch population	1.19×10^{10}	
p bunch population	2.2×10^{11}	
e norm. emittance h/v	22.0 / 22.0	μm
p norm. emittance h/v	2.5 / 2.5	μm
$e \beta_{x/y}^*$ at IP	0.20 / 0.20	m
$p \beta_{x/y}^*$ at IP	0.35 / 0.35	m
$e \sigma_{x/y}$ at IP	10.6 / 10.6	μm
$p \sigma_{x/y}$ at IP	10.7 / 10.7	μm
e / p beam current	60.1 / 1110	mA
p beam-beam $\xi_{x/y}$	0.001 / 0.001	
Peak luminosity $\mathcal{L}$	5.78×10^{33}	$\text{cm}^{-2}\text{s}^{-1}$

Hybrid EIC-LHeC Parameter Sets

Three hybrid parameter sets were developed, corresponding to: the initial hybrid at $\sim 2 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$ using unmodified EIC proton emittances; and an optimized version at $10^{34} \text{ cm}^{-2}\text{s}^{-1}$ enabled by proton emittance reduction through electron cooling and improved electron flat-beam optics. Full parameter tables for both are given in Tables 3 and 4. In both cases the ring circumference is 3834 m, the revolution frequency 78230 Hz, the electron energy 10 GeV, and the proton energy 275 GeV ($\sqrt{s} \approx 105 \text{ GeV}$).

Table 3: Hybrid EIC-LHeC Parameters, Initial Configuration ($\sim 2 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$)

Parameter	e	p	Unit
E	10.0	275.0	GeV
N	1.19×10^{10}	2.1×10^{11}	
n_{bunches}	386		
f_{collis}	3.0×10^7		Hz
Norm. ε h/v	22.0 / 22.0	3.3 / 0.3	μm
Geom. ε h/v	1.1 / 1.1	11.3 / 1.0	nm
$\beta_{x/y}^*$ at IP	1.20 / 0.05	0.12 / 0.06	m
σ_z	3.0	70.0	mm
$\sigma_{x/y}$ at IP	36.7 / 7.5	36.8 / 7.8	μm
$\sigma'_{x/y}$ at IP	30.6 / 149.9	306.3 / 130.6	μrad
$\xi_{x/y}$	3.474 / 0.679	0.001 / 0.002	
I_{beam}	57.6	1000	mA
$\sigma_{x/y}^{\text{eff}}$ at IP	52.0 / 10.8		μm
Peak $\mathcal{L}$	2.10×10^{33}		$\text{cm}^{-2}\text{s}^{-1}$

The initial hybrid achieves $\mathcal{L} \approx 2 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$ with no changes to the EIC proton emittances, satisfying the CD0 lower boundary. The proton beam-beam parameter remains safely below 0.01 in both configurations. The electron beam-beam parameter substantially exceeds unity, reflecting severe single-pass disruption; the electrons are discarded after col-

Table 4: Hybrid EIC-LHeC Parameters, Optimized Configuration ($1 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, with e -cooling and flat-beam optics)

Parameter	e	p	Unit
E	10.0	275.0	GeV
N	1.19×10^{10}	2.07×10^{11}	
n_{bunches}	386		
f_{collis}	3.02×10^7		Hz
Norm. ε h/v	22.0 / 2.2	0.9 / 0.1	μm
Geom. ε h/v	1.1 / 0.1	3.1 / 0.3	nm
$\beta_{x/y}^*$ at IP	0.25 / 0.10	0.09 / 0.05	m
σ_z	3.0	70.0	mm
$\sigma_{x/y}$ at IP	16.8 / 3.4	16.8 / 3.6	μm
$\sigma'_{x/y}$ at IP	66.8 / 32.7	182.4 / 77.8	μrad
$\xi_{x/y}$	3.474 / 6.789	0.003 / 0.007	
I_{beam}	58.0	1000	mA
$\sigma_{x/y}^{\text{eff}}$ at IP	23.8 / 5.0		μm
Peak $\mathcal{L}$	1.00×10^{34}		$\text{cm}^{-2}\text{s}^{-1}$

lision, so this does not pose a stability problem in itself, but it does raise questions about the feasibility of energy recovery for a highly disrupted bunch – a point addressed in the *Energy Recovery evaluations* section.

The optimized configuration reaches $10^{34} \text{ cm}^{-2}\text{s}^{-1}$ by reducing the proton emittances (assuming on-collision-energy electron cooling) by roughly a factor of four horizontally and three vertically relative to the initial hybrid, and by reducing the vertical electron emittance by a factor of ten via flat-beam transformation. The large proton divergences in the optimized case – up to 182 μrad horizontally – imply tight aperture requirements on the final focusing doublet, which are examined in the *IR optics evaluations* section.

BEAM-BEAM EVALUATIONS

Describe process and prompts for generation of beam-beam code and how it can be used for this case.

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Include picture of beam-beam collision

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DRAFT FOR BEAM-BEAM SECTION

The animation below demonstrates what's now possible: using Claude Code AI, we created a simulation of electron-positron beam collisions in just minutes, from a paragraph-long prompt. You can explore the full example at <https://lnkd.in/ep2pxya8>

This capability suggests a transformative shift in how students learn. Rather than passively attending lectures or running decades-old legacy codes, students could actively create simplified simulation codes, then iteratively refine

them through AI-assisted modifications to explore different physics phenomena. For instance, with the beam-beam collision example, students could progressively enhance their understanding by:

Converting to physical units

Implementing realistic beam distributions with emittance and beta-functions

Adding luminosity calculations

Investigating how luminosity depends on disruption parameters and beam offsets

IR OPTICS EVALUATIONS

Describe process and prompts for generation of IR optics, beam envelopes, estimation of the collimation depth.

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Include picture of IR optics

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IR MAGNETS EVALUATIONS

Describe process and prompts for evaluation of parameters and technological options for IR magnets, which will be determined need to be SC magnets, including discussion of Nb or Nb₃Sn choices

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ENERGY RECOVERY EVALUATIONS

Describe process and prompts for evaluation of the energy recovery

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RECIRCULATION EVALUATIONS

Describe process and prompts for evaluation of recirculation

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INITIAL COST ESTIMATIONS

Approach and Scope

The cost estimation for the hybrid EIC-LHeC electron complex was developed using AI-assisted parametric scaling from the 2018 LHeC cost study [**LHeC-costs**], which provides a detailed breakdown of the electron linac complex at CERN. The scope of the estimate covers only the new electron beam complex, the interaction region (IR), and the detector; it explicitly excludes any upgrades to the existing EIC proton beam infrastructure, since the hybrid concept assumes the RHIC-derived Hadron Storage Ring is used without major modification.

The prompts used in this step asked the AI to: identify relevant cost line items from the LHeC reference; scale each item to the EIC-LHeC parameters (principally the lower linac energy of 5–6 GeV compared to the LHeC linac); include an estimate of the required person-years (PY) of labor effort; and compare two linac configurations – a single-pass 10 GeV linac and a recirculating 5 GeV linac traversed twice to reach the same final energy. The AI was also asked to convert the resulting European cost estimate, given in MCHF, to a USA DOE cost accounting basis.

It should be noted that the cost scaling was performed with AI assistance and has not been independently verified in detail. The IR and detector cost line items were manually increased relative to the raw AI scaling output, as the automated scaling was judged to underestimate these components. The estimates should therefore be regarded as indicative and order-of-magnitude in nature, pending a more rigorous engineering cost review.

Single-Pass versus Recirculating Linac

Two configurations for the electron linac were compared:

- **Single-pass:** A 10 GeV SRF linac, through which the electron beam passes once before collision and once during energy recovery deceleration. This is the conceptually simplest configuration but requires the full SRF installed length for 10 GeV acceleration.
- **Recirculating:** A 5 GeV SRF linac (in practice assumed capable of 6 GeV) through which the beam

passes twice on the way up and twice on the way down, reaching 10 GeV (or up to 18 GeV with three passes) with half the SRF installed length. The additional cost is two recirculation arcs with conventional (non-FFA) magnets and associated transport optics.

The recirculating configuration is directly analogous to the LHeC linac-ring design and to the operating principle of CEBAF at Jefferson Lab. It reduces the dominant SRF cost roughly by half while adding a comparatively modest arc cost, yielding a lower total. A 6 GeV linac with three recirculations would cover the full EIC energy range up to 18 GeV, providing useful headroom above the 10 GeV nominal operating point.

Cost Comparison

Table 5 summarizes the cost and labor estimates for both configurations. All costs are in MCHF; PY denotes person-years of labor effort.

Table 5: Hybrid EIC-LHeC Cost and Manpower Estimate Comparison, scaled

Component	Single MCHF	Single PY	Recirc MCHF	Recirc PY
SRF System	134	–	67	–
Cryogenic Infra.	14	–	7	–
Civil Eng.	96	–	48	–
General Infra.	19	–	10	–
SRF R&D	31	–	31	–
Injector	40	–	40	–
Dump & Source	5	–	5	–
Recirc. Arcs	0	–	70	100
e-complex Subt.	339	703	278	451
IR	50	50	50	50
Detector	250	100	250	100
Total Estimated	639.5	853	577.8	601

The recirculating configuration saves approximately 62 MCHF in direct costs and 252 PY in labor relative to the single-pass design, driven primarily by the halving of the SRF system, civil engineering, and cryogenic infrastructure costs. The recirculation arc additions (70 MCHF, 100 PY) recover only a fraction of these savings. On this basis the recirculating 6 GeV linac is the preferred configuration and is adopted as the reference for the DOE cost accounting in the following section.

USA DOE Cost Accounting

The recirculating configuration cost was converted to USA DOE accounting conventions, as detailed in Table 6. The conversion applies: a CHF-to-USD exchange rate of 1.25 (verified against market rates at the time of the study); fully-loaded labor at 300k USD per person-year; and a 60% contingency on the pre-contingency total, reflecting the very preliminary and unverified nature of this estimate.

Table 6: USA DOE Cost Accounting for EIC-LHeC Hybrid (Recirculating 6 GeV Linac Configuration)

Item	Value (MUSD)
Material Costs (MCHF $\times$ 1.25)	722.2
Labor Costs (601 PY $\times$ 300k)	180.5
Pre-Contingency Total	902.7
Contingency (60%)	541.6
Total with Contingency	1444.3

The total estimated project cost of ~ 1.4 B USD is approximately half the upper bound of the CD0-approved cost range for the full EIC project. While this estimate is highly tentative, the factor-of-two margin suggests that the hybrid EIC-LHeC concept warrants more detailed evaluation. The large 60% contingency applied here is intentional: at this stage of conceptual design, informed by AI-assisted scaling rather than bottom-up engineering estimates, a conservative contingency is appropriate. A more rigorous cost review, with independent validation of each line item, would be a natural next step before drawing stronger conclusions.

ADVANCED COST ESTIMATIONS

Describe process and prompts for advanced evaluation of the cost of EIC-LHeC accelerator

Will be written by Carl Andreas Lindstrøm later in April

This will be based on the cost modelling described both in the ESSPU HALHF backup document (<https://arxiv.org/pdf/2503.23489>) as well as an IPAC 2025 paper (<https://arxiv.org/pdf/2505.21654>). It will also reference the code via the following link to the first, more crude cost optimization code (<https://github.com/carlandreaslindstrom/ColliderCostModel>); and to the up-to-date cost optimization code used in ABEL (https://github.com/abel-framework/ABEL/blob/main/abel/classes/cost_model.py).

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REQUIRED R&D AND NEXT STEPS

The hybrid EIC-LHeC concept rests on several assumptions that require dedicated R&D before the design can be considered credible at project-proposal level. The principal open questions, identified in dialogue with AI, are:

1. **High-current ERL operation.** Energy recovery efficiency of $\lesssim 10^{-4}$ fractional loss is required at 58 mA and 10 GeV. A staged program – low-current/high-energy at CEBAF, high-current/low-energy at PERLE – would bound the two critical parameters.

2. **Polarized electron source current.** The 58 mA design value far exceeds the present state of the art (~ 1 mA CW). Multiplexing N sources via RF beam mergers is a plausible path once 10 mA from a single source is demonstrated at Jefferson Lab.
3. **IR final doublet and collimation.** The small proton β^* (0.09–0.12 m) drives large IP divergences (~ 300 μ rad), requiring large-aperture SC quadrupoles and tight collimation, with potential impact on 4π detector coverage and MDI compatibility with ePIC.
4. **Intrabeam scattering.** Higher proton bunch charge (factor-of-three increase) raises the IBS rate; electron cooling must overcome IBS growth to maintain the emittances assumed in the 10^{34} optimized configuration.
5. **Energy recovery of disrupted beam.** Electron beam-beam parameters $\xi_y^e \approx 6.8$ imply severe single-pass disruption. Whether the resulting phase-space volume fits within return-arc and ERL deceleration acceptance requires dedicated simulation.
6. **ePIC detector compatibility.** Any change to the proton FD optics or electron beam delivery must preserve the existing MDI geometry, solenoid compensation, and synchrotron radiation masking. Early engagement with the ePIC detector group is essential.

Logical next steps are: (1) systematic beam-beam parameter scans with strong-strong or weak-strong codes; (2) conceptual IR lattice and MDI assessment; (3) ERL lattice design and disrupted-beam acceptance study; (4) IBS/cooling power modeling; (5) bottom-up cost re-estimation; and (6) programmatic compatibility assessment with the EIC CD schedule. The hybrid design illustrates how AI-assisted parametric exploration can rapidly map an alternative design space and surface the critical questions for a more detailed study – accelerating the early conceptual phase without replacing rigorous engineering analysis.

VERIFICATION OF AI RESULTS

Discuss how verification of AI results will be performed, and how it is connected with our believe that students will learn accelerator physics from such educational approach.

DRAFT FOR EVALUATION SECTION:

For instance, referring to the above mentioned beam-beam collision example, students could progressively enhance their understanding by making simulation more and more realistic, but most importantly...

Critically, students would develop diagnostic tests to verify that physics principles are correctly implemented – learning to validate AI-generated code against their understanding of the underlying physics.

This approach ensures that as we leverage AI tools, we continue developing human experts capable of designing and building the accelerators and colliders of tomorrow. The

goal isn't to replace human expertise, but to accelerate its development while maintaining rigor and deep understanding.

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CONNECTION TO GENESIS PROGRAM

Discuss connection to Genesis program.

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SELF-ASSESSMENT AND NEXT STEPS

Discuss how we feel about the described approach, what is missing, where we are not sure, what are directions of future developments.

Text

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CONCLUSION

Write conclusion

Text

Text

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